

Creon®**10 000**

150 mg Pancreatin
(pancreas powder)

hard gelatine capsule filled with gastro-resistant granules

Creon® 10,000 is bicoloured capsule with brown opaque cap and transparent body of hard gelatin capsules filled with gastro-resistant granules (Creon Minimicrospheres™) for oral administration. 1 capsule of Creon® 10,000 contains 150 mg Pancreatin (pancreas powder) corresponding to

Amylase	8 000 Ph.Eur. units (=BP units)
Lipase	10 000 Ph.Eur. units (=BP units)
Proteas	600 Ph.Eur. units (=BP units)

Produced from porcine pancreatic tissue.

Excipients: macrogol 4000, hypromellose phthalate, dimethicone 1000, triethyl citrate, cetyl alcohol, iron oxide (E172), titanium dioxide (E171), sodium lauryl sulphate, gelatin.

Indications

Treatment of pancreatic exocrine insufficiency in paediatric and adult patients often associated with, but not limited to:

- cystic fibrosis
- chronic pancreatitis
- pancreatic surgery
- gastrectomy
- pancreatic cancer
- gastrointestinal bypass surgery (e.g. Billroth II gastroenterostomy)
- ductal obstruction of the pancreas or common bile duct (e.g. from neoplasm)
- Shwachman-Diamond Syndrome
- Creon® 10,000 only: For the improvement of digestion and relief of maldigestion symptoms after consumption of a (fatty) meal; like sensation of heaviness, bloating, abdominal distension, abdominal pain/cramping and/or diarrhoea.

Dosage and Administration

The posology aims at individual needs and depends on the severity of the disease and the composition of food. It is recommended to take the enzymes during or immediately after the meals.

The capsules should be swallowed intact, without crushing or chewing, with enough fluid during meals or snacks.

When swallowing of capsules is difficult (e.g. small children or elderly patients), the capsules may be carefully opened and the pellets added to acidic soft food [pH < 5.5] that does not require chewing, or the pellets will be taken with acidic liquid with [pH < 5.5]. This could be apple sauce or yogurt or fruit juice with a pH less than 5.5, e.g. apple, orange or pineapple juice. This mixture should not be stored. Crushing and chewing of the pellets or mixing with food or fluid with a pH greater than 5.5 can disrupt the protective enteric coating. This can result in early release of enzymes in the oral cavity and may lead to reduced efficacy and irritation of the mucous membranes.

Care should be taken that no product is retained in the mouth.

It is important to ensure adequate hydration at all times, especially during periods of increased loss of fluids. Inadequate hydration may aggravate constipation. Any mixture of the pellets with food or liquids should be used immediately and should not be stored.

Dosing in pediatric and adult patients with cystic fibrosis

Based upon a recommendation of the Cystic Fibrosis Consensus Conference, the US CF Foundation case-control study, and the UK case-control study, the following general dosage recommendation for pancreatic enzyme replacement therapy can be proposed:

- Weight-based enzyme dosing should begin with 1 000 lipase units/kg/meal for children less than four years of age and with 500 lipase units/kg/meal for those over age four.
- Dosage should be adjusted according to the severity of the disease, control of steatorrhea and maintenance of good nutritional status.
- Most patients should remain below or should not exceed 10 000 lipase units/kg body weight per day or 4000 lipase units/gram fat intake

Dosing in other condition associated with exocrine pancreatic insufficiency

Dosage should be individualized by patient according to the degree of maldigestion and the fat content of the meal. The required dose for a meal ranges from about 25 000 to 80 000 Ph. Eur. units of lipase and half of the individual dose for snacks.

Dose of 1 to 2 capsules of Creon® 10,000 per meal for the improvement of digestion and relief of maldigestion symptoms after consumption of a (fatty) meal (Refer to section Indications).

Contraindications

Hypersensitivity to the active substance or to any of the excipients.

Warnings and special precautions for use

Strictures of the ileo-caecum and large bowel (fibrosing colonopathy) have been reported in patients with cystic fibrosis taking high doses of pancreatin preparations. As a precaution, unusual abdominal symptoms or changes in abdominal symptoms should be medically assessed to exclude the possibility of fibrosing colonopathy, especially if the patient is taking in excess of 10000 units of lipase/kg/day.

Interaction with other medicinal

No interaction studies have been performed.

Pregnancy and lactation

Pregnancy

For pancreatic enzymes no clinical data on exposed pregnancies are available.

Animal Studies show no evidence for any absorption of porcine pancreatic enzymes. Therefore, no reproductive or developmental toxicity is to be expected.

Caution should be exercised when prescribing to pregnant women.

Lactation

No effects on the suckling child are anticipated since animal studies suggest no systemic exposure of the breastfeeding woman to pancreatic enzymes. Pancreatic enzymes can be used during breastfeeding.

If required during pregnancy and lactation Creon® should be used in doses sufficient to provide adequate nutritional status.

Effects on ability to drive and use machines

Creon® has no or negligible influence on the ability to drive and use machines.

Undesirable effects

In clinical trials more than 1000 patients were exposed to Creon®.

The most common reported adverse reactions were gastrointestinal disorders which were mostly mild or moderate in severity.

The following adverse reactions have been observed during clinical trials with the below indicated frequencies.

Organ system	Very common ≥ 1/10	Common ≥ 1/100 to < 1/10	Uncommon ≥ 1/1000 to < 1/100	Frequency not known
Gastrointestinal disorders	abdominal pain*	nausea, vomiting, constipation, abdominal distention, diarrhea*		strictures of the ileo-caecum and large bowel (fibrosing colonopathy)
Skin and subcutaneous tissue disorders			rash	pruritus, urticaria
Immune system disorders				hypersensitivity (anaphylactic reactions).

*Gastrointestinal disorders are mainly associated with the underlying disease. Similar or lower incidences compared to placebo were reported for abdominal pain and diarrhea.

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Allergic reactions mainly but not exclusively limited to the skin have been observed and been additionally identified as adverse reactions during post-approval use.

Because these reactions were reported spontaneously from a population of uncertain size, it is not possible to reliably estimate their frequency.

Paediatric population

No specific side effects were found. In children with cystic fibrosis frequency, type and severity of side effects were similar to those in adults.

Overdose

Extremely high doses of pancreatin have been reported to associated with hyperuricosuria and hyperuricaemia.

Pharmacodynamics

Pharmacotherapeutic group: Multienzymes (amylase, lipase, protease).

ATC code: A09A A02

Creon[®] contains porcine pancreatin formulated as enteric-coated (acid-resistant) pellets within gelatine capsules.

The capsules dissolve rapidly in the stomach releasing plenty of pellets, a multi-dose principle which is designed to achieve good mixing with the chyme, emptying from the stomach together with the chyme and after release, good distribution of enzymes within the chyme.

When the pellets reach the small intestine the coating rapidly disintegrates (at pH > 5.5) to release enzymes with lipolytic, amylolytic and proteolytic activity to ensure the digestion of fats, starches and proteins. The products of pancreatic digestion are then either absorbed directly, or following further hydrolysis by intestinal enzymes.

Clinical efficacy:

Overall 30 studies investigating the efficacy of Creon® in patients with pancreatic exocrine insufficiency have been conducted. Ten of these were either placebo-controlled studies performed in patients with cystic fibrosis, chronic pancreatitis or post-surgical conditions.

In all randomized, placebo-controlled, efficacy studies, the pre-defined primary objective was to show superiority of Creon® over placebo on the primary efficacy parameter, the coefficient of fat absorption (CFA).

The coefficient of fat absorption determines the percentage of fat that is absorbed into the body taking into account fat intake and fecal fat excretion. In the placebo-controlled PEI studies, the mean CFA (%) was higher with Creon® treatment (83.0%) as compared to placebo (62.6%). In all studies, irrespective of the design, the mean CFA (%) at the end of the treatment period with Creon® was similar to the mean CFA values for Creon® in the placebo-controlled studies.

Treatment with Creon® markedly improves the symptoms of pancreatic exocrine insufficiency including stool consistency, abdominal pain, flatulence and stool frequency, independent of the underlying disease.

Paediatric population:

In cystic fibrosis (CF) the efficacy of Creon® was demonstrated in 288 paediatric patients covering an age range from newborns to adolescents. In all studies, the mean end-of-treatment CFA values exceeded 80% on Creon® comparably in all paediatric age groups.

Pharmacokinetics

Animal studies showed no evidence for absorption of intact enzymes and therefore classical pharmacokinetic studies have not been performed. Pancreatic enzyme supplements do not require absorption to exert their effects.

On the contrary, their full therapeutic activity is exerted from within the lumen of the gastrointestinal tract. Furthermore, they are proteins, and as such undergo proteolytic digestion while passing along the gastrointestinal tract before being absorbed as peptides and amino acids.

Preclinical Safety Data

Preclinical data show no relevant acute, subchronic or chronic toxicity. Studies on genotoxicity, carcinogenicity or toxicity to reproduction have not been performed.

Incompatibilities

Not applicable.

Shelf life and storage conditions

2 years.

Do not store above 30°C.

Storage and disposal

Do not use the medicine after the expiry date stated on the carton.
Keep this medicine out of the reach and sight of children.

Pack size(s)

100 capsules in blister pack.

Further information

The information in this leaflet is limited.
For further information, please contact your doctor or pharmacist.
Any unused product or waste material should be disposed of accordance with the local requirements.

Manufactured by

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